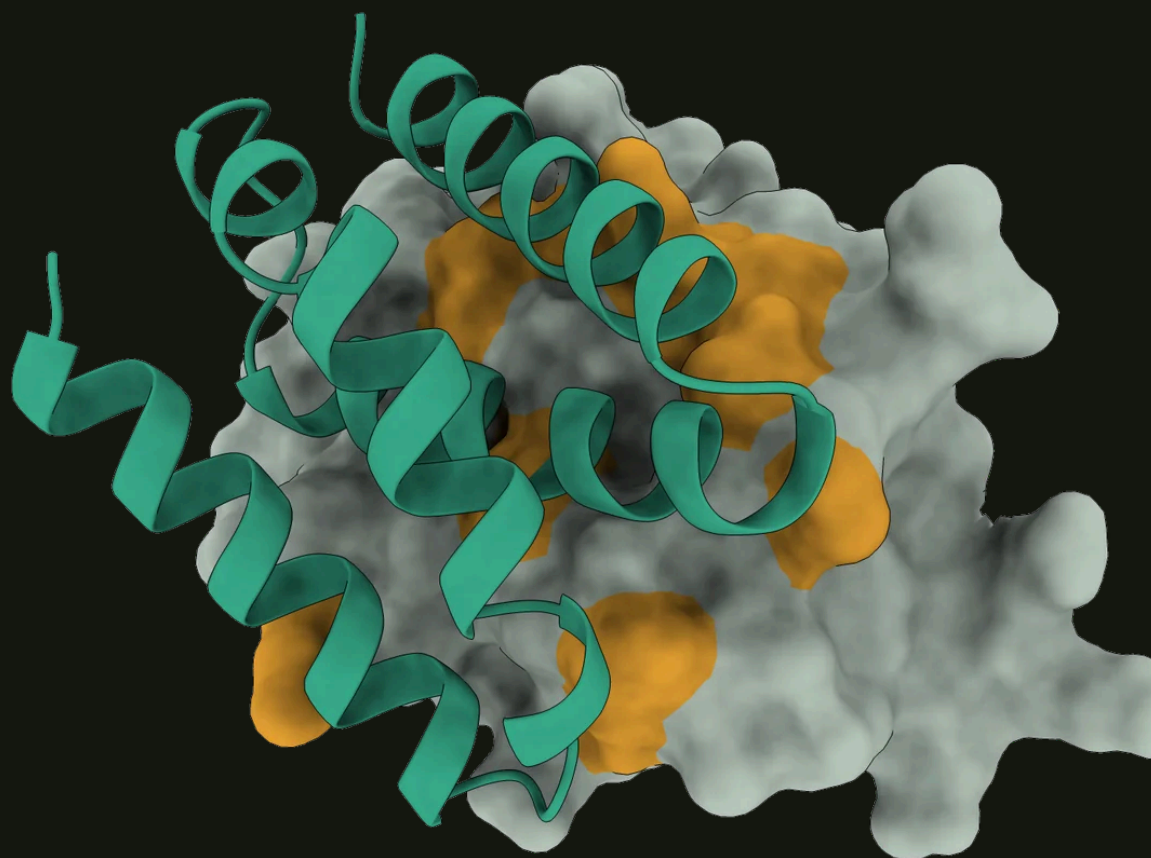


You already know your target.

“Design binders against PD-L1 to block its interaction with PD-1.”

No structure named, no epitope chosen. It ranked the nine solved structures of PD-L1 by their measured interfaces, proved the chain it picked was PD-L1 and not the nanobody bound to it, chose a nine-residue epitope, sized the campaign from its own trial, and returned **715 gated candidates** from 5,824 refolds — **\$2.12** of model spend and 26.4 GPU-hours, unattended.



Nine solved structures. Ranked on the interface, not the citation count.

ENTRY	Å	BSA Å ²	IFACE RES	H - BONDS	PHOB
7CZD anti-PD-L1 VHH	1.64	2,449	44	30	0.34
8AOK VHH 6	1.60	2,215	46	37	0.22
8ZNL PD-L1 de novo binder	1.77	1,844	31	17	0.39
5JDS Nanobody	1.70	1,648	30	23	0.30
8AOM VHH6	2.20	1,307	17	3	0.71

WHY NOT THE RUNNER-UP

8AOK — second-largest BSA and most H-bonds, but low hydrophobic fraction (0.22) suggests a flatter, more polar epitope, harder to reproduce with a de novo binder

Every figure in that table is computed from coordinates before any model reads the list, so the choice is argued over measurements rather than over abstracts — 7CZD on the largest, best-resolved description of the epitope every characterised PD-L1 blocker uses.

Chain A of this structure is not the target. It is the reagent.

7CZD holds PD-L1 on chain B and the anti-PD-L1 VHH used to crystallise it on chain A. Told to design against “PD-L1”, an earlier version of this stage picked chain A and put its hotspots on the nanobody’s own CDR loop.

100%

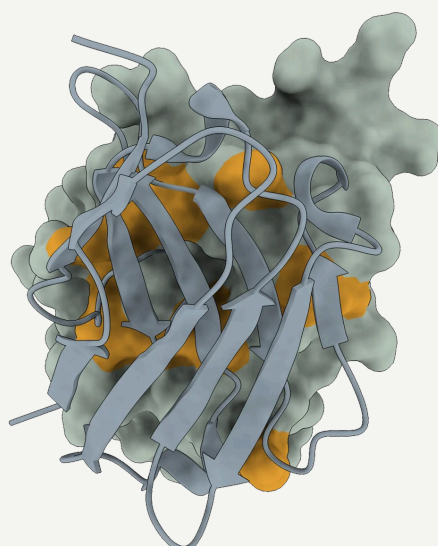
CHAIN B VS Q9NZQ7

20%

CHAIN A VS Q9NZQ7

WHAT RUNS NOW, BEFORE A TRIM IS BUILT

Every modelled residue of the chosen chain is aligned against the target’s canonical UniProt sequence, and a mismatch halts the run. Checking the residues cannot catch this — they are real and correctly numbered, just on the wrong protein.



The nanobody bound to PD-L1 in 7CZD, down the epitope axis. Designing against its loops would have been geometrically unremarkable and biologically worthless.

Nine residues, on the face PD-1 itself binds.

9

HOTSPOTS DECLARED

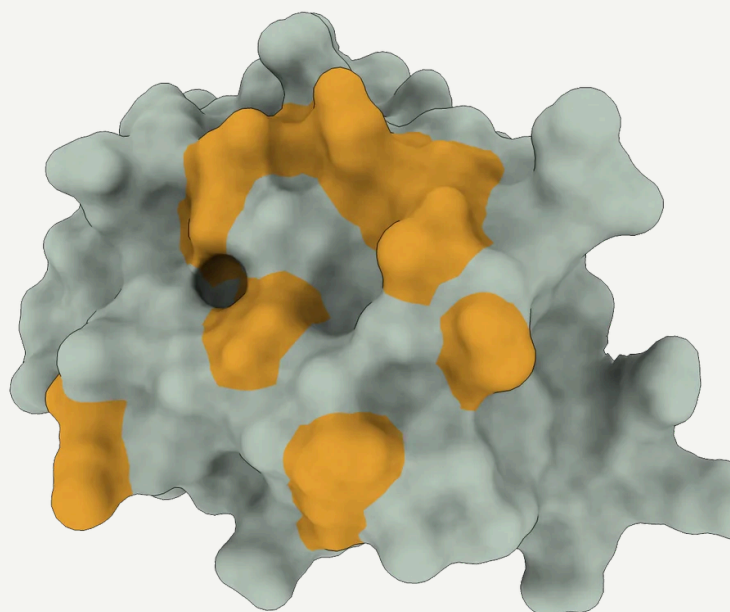
2

CANDIDATE REGIONS
SCORED

excellent

60% HYDROPHOBIC

Tyr56, Glu58, His69, Lys75, Arg113, Gly119, Ala121, Asp122, Tyr123 — the C-terminal F/G-strand, chosen over one alternative face. Tyr56 (-3.9), His69 (-2.9) carry a computed $\Delta\Delta G$. A region declares at most twelve: more is not stricter, because RFD3's hit rate falls as the set grows.

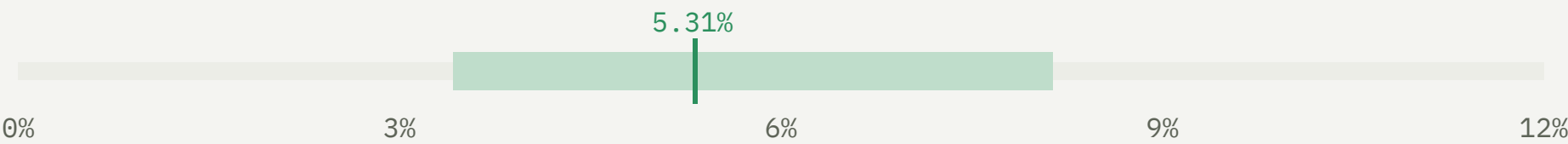


The epitope with no binder present. Residue names and sidechain atoms are read from the file at each position — a hotspot table can be internally consistent and still describe a different structure of the same protein.

The trim then did nothing, which is also a result: at 117 residues against a 220 budget the chain was kept whole, in 1 segment (70-86, /0, B18-134). Deciding a target already fits is as much that stage's job as cutting one down.

It measured its own hit rate before spending a day of GPU.

A 580-backbone trial, refolded 4 sequences deep: **19 of 358** backbones produced a design clearing every gate at the excellence bar.



95% Wilson interval 3.42–8.14%. At this rate a trial sees a handful of hits, and a point estimate off a handful is not a number worth four days — so the campaign is sized against the pessimistic end.

BASIS	BACKBONES	REFOLDS	GPU-HOURS
point estimate	1,526	3,768	11.5
Wilson lower bound	2,366	5,842	17.8

VERDICT · SCALE UP

It also raised its own success bar from 0.7 to **0.85**. An unusually good target should not be sized to the same fixed bar as a marginal one, so the bar climbs to the strictest rung the trial still supports and the budget still affords.

Eight gates, each applied to every refold on its own.

2,364

BACKBONES DIFFUSED

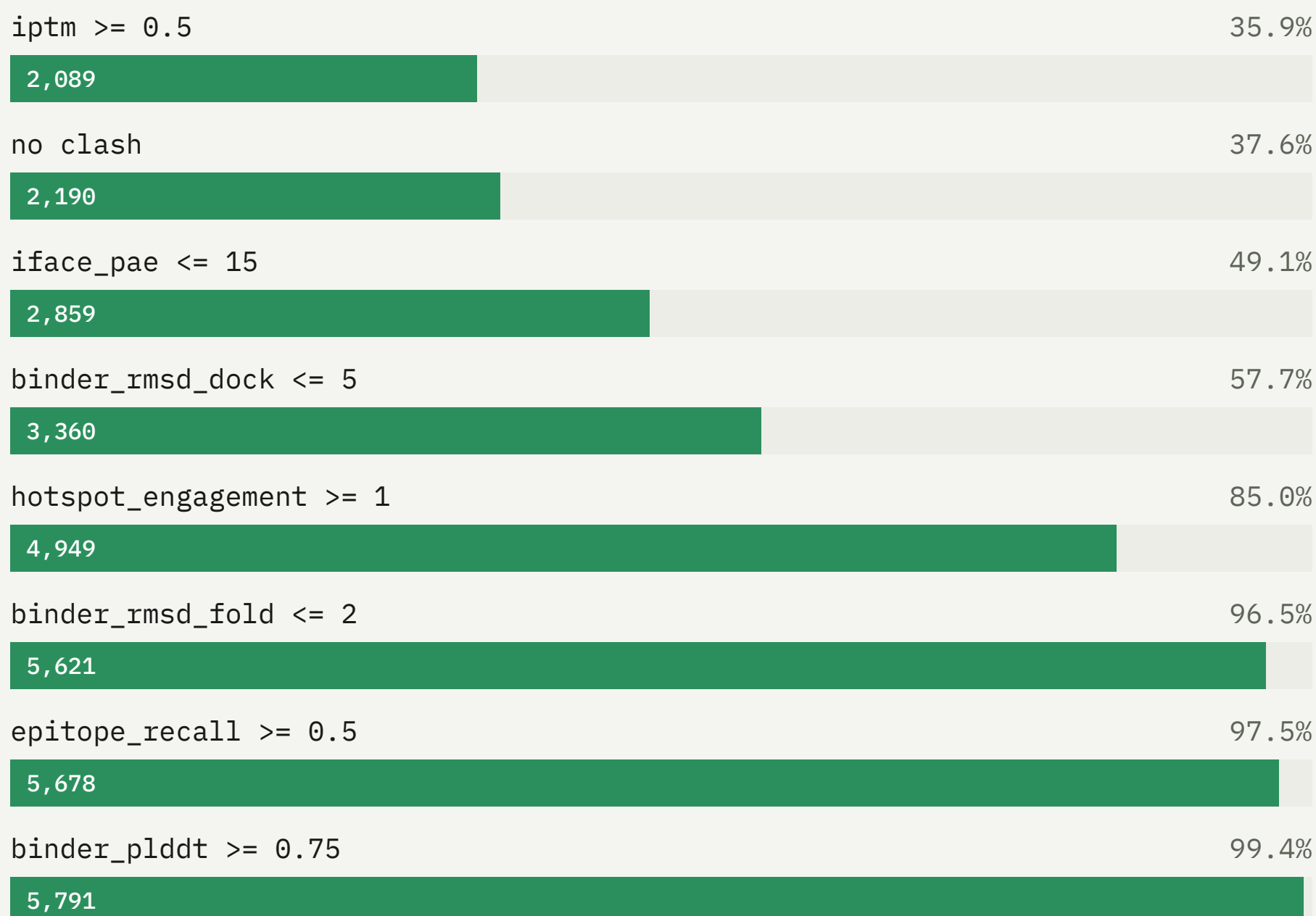
5,824

REFOLDS SCORED

715

SURVIVORS

A cheap sidecar prefilter kept 62% of the backbones (1,456), and each survivor was folded as 4 sequences. Bars are the share of all 5,824 refolds each gate would pass *alone*; the 715 survivors are the intersection, across 462 distinct backbones.



A model can be confident about an interface it invented.

The refolder is never shown the docked pose, so ipTM reports confidence in whatever interface the model chose — not in the one that was designed for. Rank on it alone and you select confidently mis-docked binders.

1,026

REFOLDS OVER IPTM 0.7

95%

OF THOSE DOCKED ≤ 5 Å

51

CONFIDENT AND MIS-DOCKED

THIS TARGET IS THE EASY CASE

95% is unusually good. Of designs over ipTM 0.7 on two other campaigns in this repo, only **45%** (8TAC) and **8%** (CD79b) were docked on target at all. The dock-RMSD gate is the one doing the work, and on a harder target it does nearly all of it.

The commonest single failure was geometric and it was rarely alone: 2,464 refolds failed dock-RMSD first, and 2,232 of those (90.6%) were below the ipTM gate as well.

THE CHECK THE RUN DID NOT DO

It aimed at PD-1's own footprint, without ever being shown it.

Nothing in the campaign opened 4ZQK, the PD-1/PD-L1 complex. Superposed onto the top-ranked design afterwards (0.83 Å over 115 Cα, 99% identity), the two binders land on the same face.

14/18

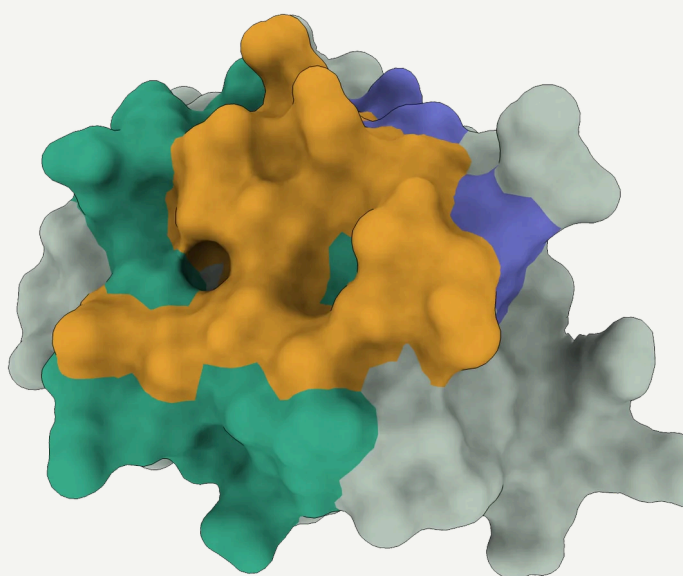
PD-1 CONTACTS ALSO
TOUCHED

78%

FOOTPRINT OVERLAP

7/9

HOTSPOTS ARE PD-1
CONTACTS



■ both (14) ■ PD-1 only (4) ■ design only (12) ■ neither

Manual analysis, not part of the run — a UCSF ChimeraX session done by hand afterwards, at a 4.5 Å cutoff. An unversioned session cannot be extracted, so it is not presented as something the pipeline produced.

RANK 1

84 residues, ipTM 0.887, dock-RMSD 1.16 Å, binder pLDDT 0.82, Rosetta $\Delta\Delta G$ -66 REU. One of 20 shortlisted, each from a different backbone.

THE BILL

Three reasoning stages. Five model calls. The rest is arithmetic.

\$2.12

MODEL SPEND, CAP \$10

26.4

GPU-HOURS, ONE CARD

27

HOURS WALL CLOCK

STAGE	MODEL CALLS	GPU-H
target intel — structures ranked, chain verified	1	—
interface — the epitope	3	—
trim, spec, pilot	0	0.87
calibration — the trial and its verdict	0	4.84
production, gating, ranking	0	20.65
summary — the write-up	1	—

TWO OF THOSE FIVE CALLS WERE A BUG

The interface stage ran 3 times because the query it was handed never named the entry the orchestrator had *already downloaded*: given a description and no accession, the skill twice concluded no structure existed and asked for one. The query now always states the id and chains. This run also drove `claude-sonnet-5` (\$2.11 of the \$2.12); the default is now `gemini-3.7-flash`, about four times cheaper on input — and no model is built in, since Claude, Gemini and GPT each drive the same loop, tools and cost ledger.

Nothing here has been near a bench.

Every figure on this deck is computational: folding confidence, interface geometry, buried surface area, Rosetta terms over 300 relaxed poses. Designs like these need expression, purification and a binding assay before any of it is a result. The trial, the gates and the ranking exist to make that shortlist small and defensible — not to skip it.

And the code has moved since the run.

This campaign ran 2026-08-21–2026-08-22. One gate has been deliberately loosened since: `hotspot_engagement` went from 1.0 to 0.75, because requiring every hotspot rejected refolds for missing residues their own backbone never targeted. Re-gating the same 5,824 refolds through today’s code:

	SURVIVORS	BACKBONES	TOP 20
as it ran, 2026-08-22	715	462	—
re-gated today	752	488	same 20

Same twenty designs, in the same order but for one adjacent swap, and the trial’s verdict and self-raised bar are unchanged. The figures above are the run’s own, at the threshold in force when it ran — the shortlist shown on this deck came out of that gate, so mixing the two would be worse than stating both.

Free for any non-commercial use.

Source-available under PolyForm
Noncommercial 1.0.0. Academic and
personal use, no fee, no negotiation.
Commercial use needs a separate licence.

Entry point shown here	<code>--workflow binder, target named</code>
The other one	<code>--workflow ppi, one sentence, no target</code>
And a third	<code>--workflow structure, your own file</code>
This campaign, written up in full	<code>campaign_pdl1.html</code>

Four campaigns documented end to end, every figure extracted from the
run that produced it — including the ones that make the tool look worse.